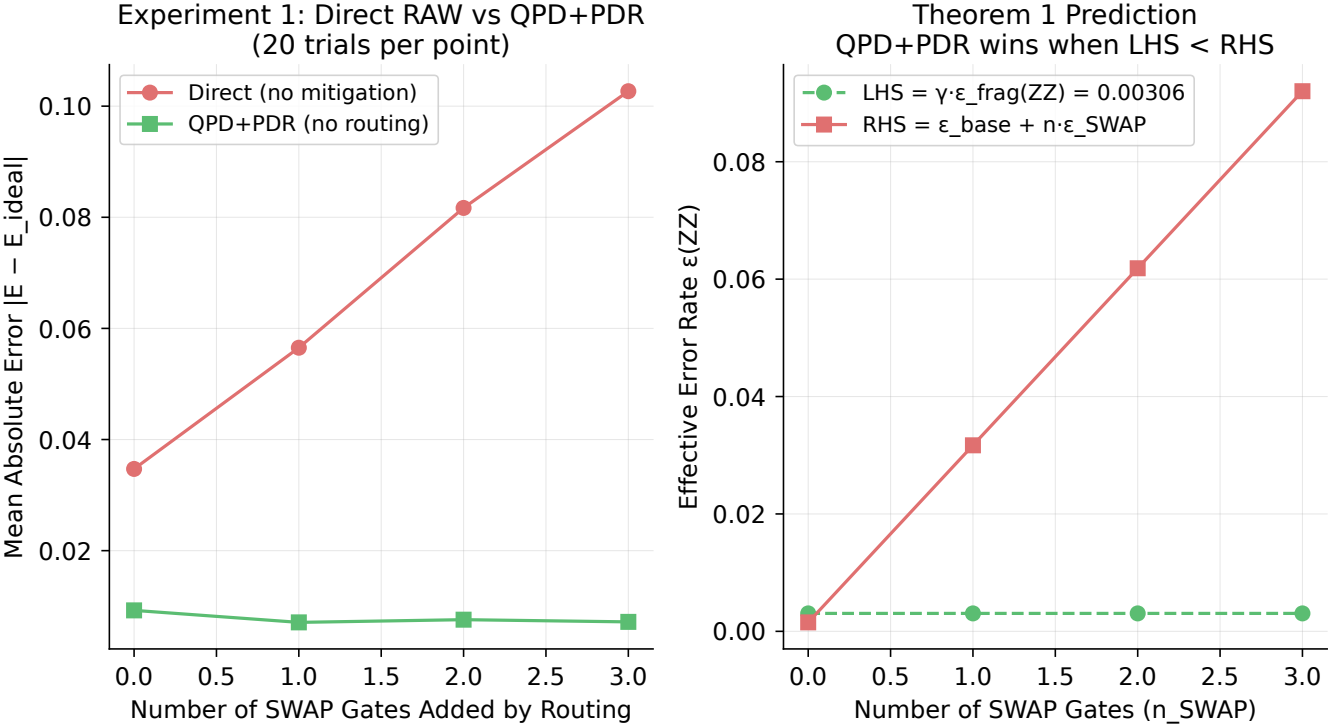
Experiment 1: Direct RAW vs QPD+PDR
(20 trials per point)
Direct (no mitigation)
QPD+PDR (no routing)
Mean Absolute Error |E − E_ideal|
0.10
0.08
0.06
0.04
0.02
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Number of SWAP Gates Added by Routing
Theorem 1 Prediction
QPD+PDR wins when LHS < RHS
LHS = γ·ε_frag(ZZ) = 0.00306
RHS = ε_base + n·ε_SWAP
Effective Error Rate ε(ZZ)
0.08
0.06
0.04
0.02
0.00
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Number of SWAP Gates (n_SWAP)